

Data Science Tools And Software

Dr. Laila M. Fatehy

Department Of Mathematics And Computer Science
Alexandria University

Data Visualization



Data Visualization

Data visualization is the process of using visual elements like charts, graphs, or maps to represent data. It translates complex, high-volume, or numerical data into a visual representation that is easier to process.

Importance of Charts and Graphs in Data Visualization

Charts and Graphs help to:

- **Identify trends and patterns:** Data points are observed together and compared to see the increase/decrease in their data values with respect to specific parameters and help to identify trends & patterns in the values.
- **Approximately predict values at future points:** Multiple data points in a chart are observed together and give a predictable correlational value for the next data point.
- **Identify deviation:** How much the value deviates from the expected value.
- **Compare between significant points:** Two or more significant data points and their values can be compared to one another easily visually due to them being plotted in the same chart.

Understanding Your Data and Visualization Goals

Before choosing the right chart it's essential to understand the nature of your data and the message you want to convey. To understand consider the following questions:

1. What is the primary purpose of the visualization?

Are you comparing values, showing trends over time, illustrating relationships, or displaying the composition of data?

2. What type of data are you working with?

Is it categorical, numerical, or a mix of both?

3. How many variables are you visualizing?

Are you dealing with a single variable, two variables, or multiple variables?

4. What is the volume of data points?

Do you have a few data points or a large dataset?

Visualization Technique – Selecting Chart Types

Different charts serve different purposes and can highlight various aspects of your data.

Common roles for data visualization include:

- showing change over time
- showing a part-to-whole composition
- looking at how data is distributed
- comparing values between groups
- observing relationships between variables
- looking at geographical data

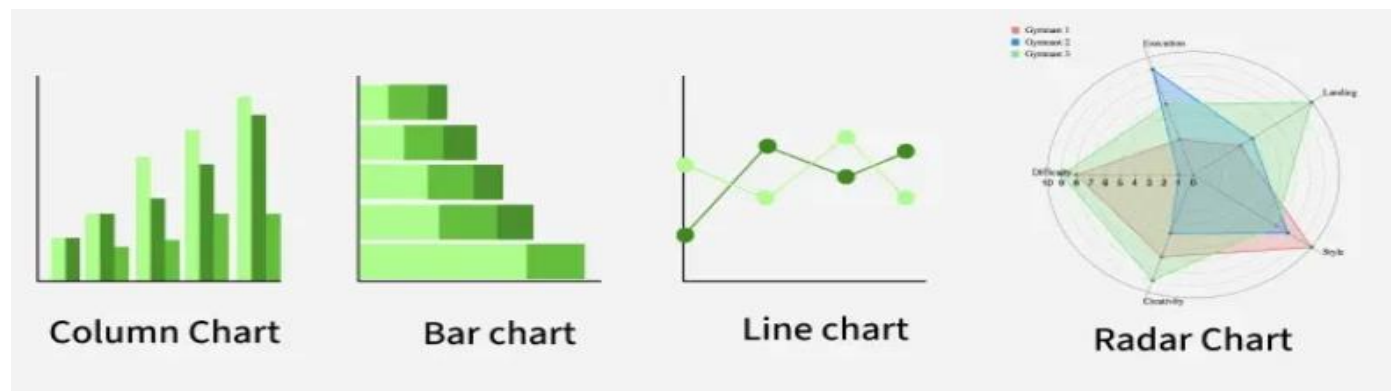
Selecting Chart Types

1. Comparison Charts

- Comparison charts are used to show how different things are related or how they change.
- They help us see patterns trends and differences in data more easily.
- For example: **Compare** sales numbers for two months or see how one product performs against others. **Compare** product sales or city populations. **Compare** multiple variables across different categories

Charts used for comparison are:

Column Chart, Bullet Graph, Mekko Chart, Pie Chart, Bar Graph, Line Graph, Scatter Plot



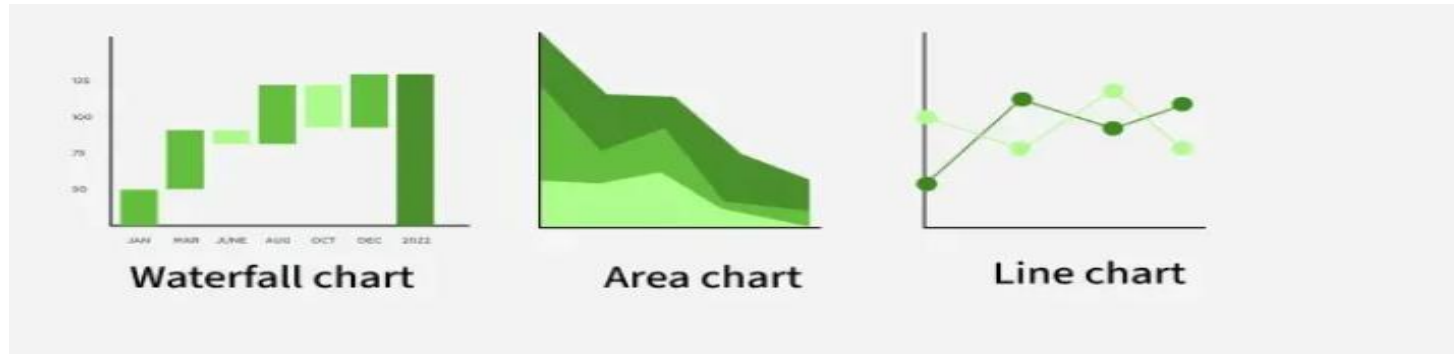
Selecting Chart Types

2. Trend Charts

- Trend charts are used to show how data changes over time.
- They visually represent the movement of variables helping to identify patterns and trends.
- For example: data points across a timeline let you understand how something evolves like *stock prices* or *temperature changes*.

Charts used are:

Line Graph, Dual-Axis Line Graph, Column Chart



Selecting Chart Types

3. Relationship Charts

- Relationship charts are used to show a connection or correlation between two or more variables.
- By visually representing the interdependencies and associations relationship charts provide a clear way to understand the relationship between the variables.

Charts used for comparison are:

Scatter Plot, Bubble Chart, Line Graph



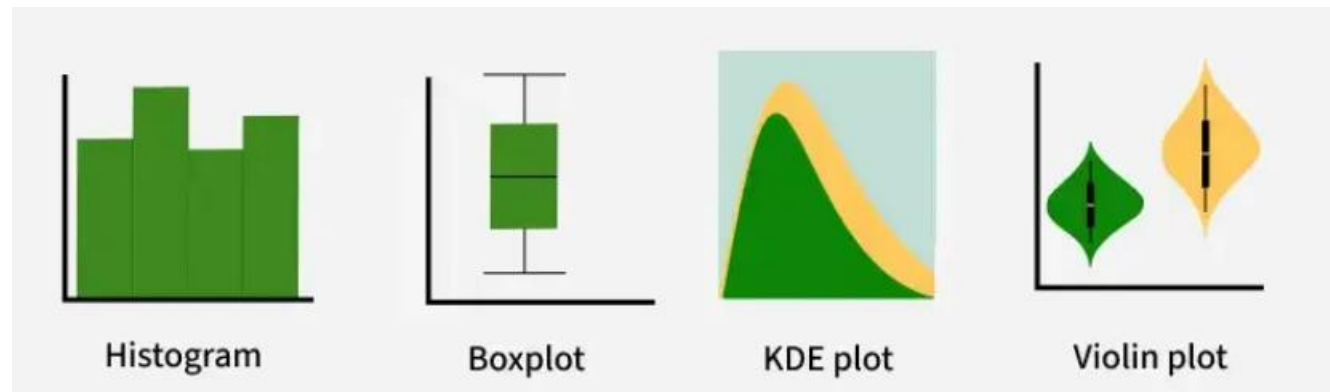
Selecting Chart Types

4. Distribution Charts

- Distribution charts help you to understand outliers, the normal tendency, and the range of information in your values.
- By showing where most data points fall distribution charts make it easier to understand patterns and spot anything that stands out.

Charts used for comparison are:

Scatter Plot, Mekko Chart, Line Graph, Column Chart, Bar Chart



Selecting Chart Types

5. Composition Charts (part-to-whole)

- Composition charts are helpful when you want to break down something into smaller parts and see how each part fits into the bigger picture.
- For example: imagine you're looking at a company's revenue over time. It can show you how different products or services contribute to the total revenue and how those contributions change each year or month.

Charts used for comparison are:

Stacked Bar Graph, Mekko Chart, Stacked Column Chart, Stacked Bar Graph, Area Chart, Waterfall Chart



Selecting Chart Types

6. Geographical Charts

- When working with geographical data we often need charts that can represent regions and locations in a meaningful way.
- Choropleths and cartograms are two charts designed specifically for this purpose.

Charts used for comparison are:

Choropleths, Cartograms



Choosing a visualization-Examples

One-Dimensional Data

A **one-dimensional visualization** only visualizes a single feature of the dataset. *For example:*

- "I want to know the proportion of **people who have cars** in my data“
- "I want to know how many of each **product type** are in my data"

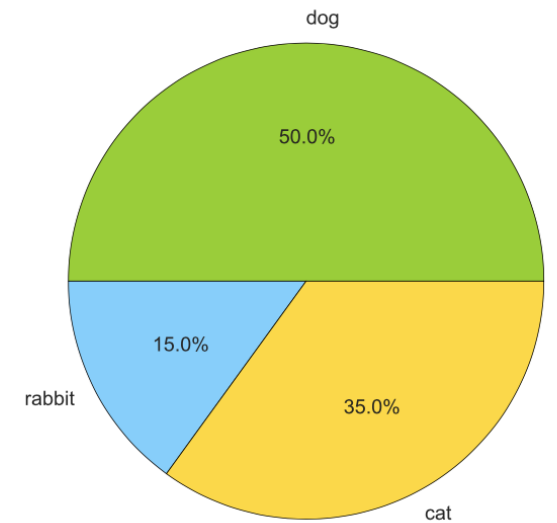
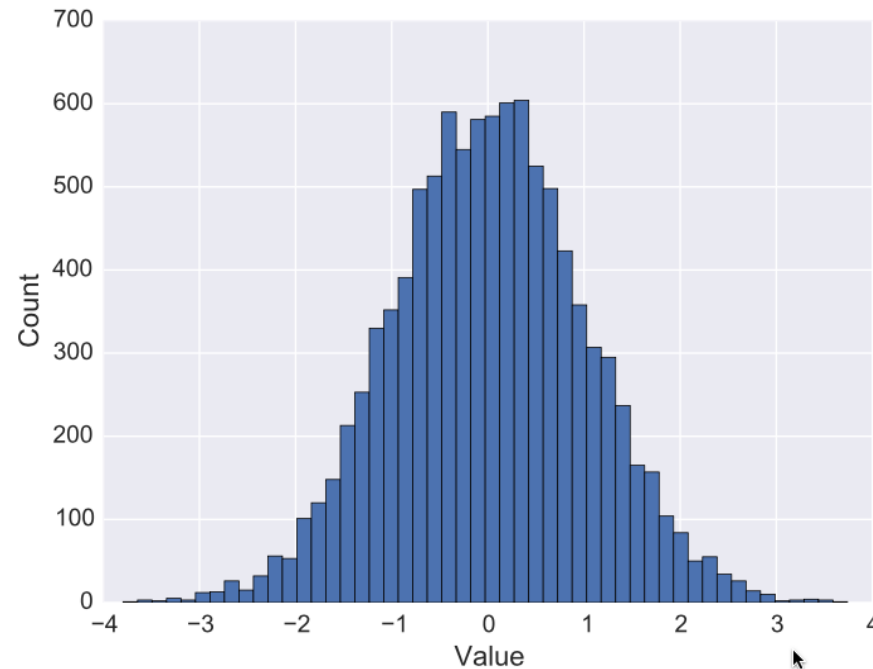
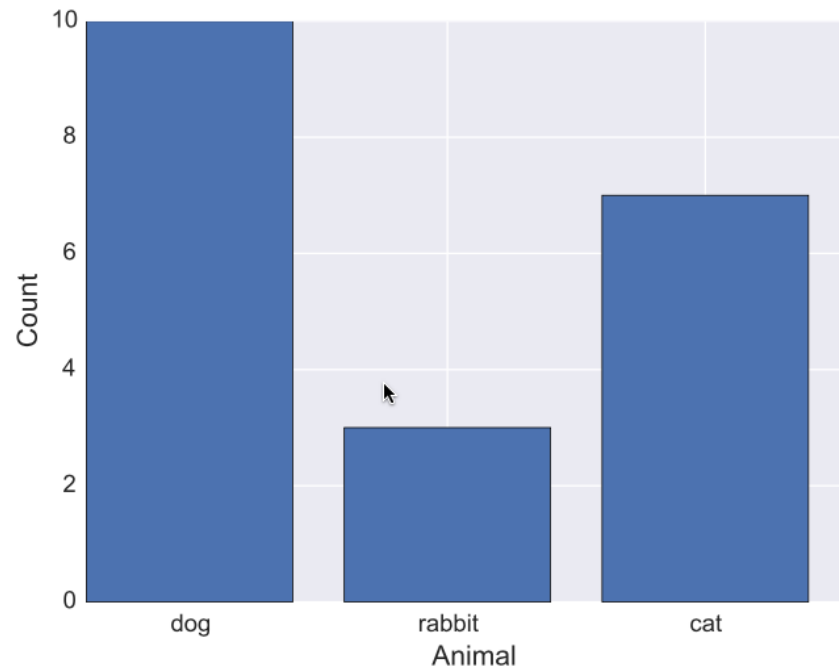
To visualize one-dimensional data, use a **histogram** or a **pie chart**.

Histograms Show Counts

- For categorical or ordinal data, show counts for each type of data using bars (length = count).
- For numerical data, use **bucketing** across a distribution.
- A **histogram** shows you the **shape** and the **spread** of numerical data.

Pie Charts Show Percentages

A **pie chart** shows the proportion of the data that falls into each category of the feature. The proportions should always add up to 100%.



matplotlib and seaborn **Make Visualizations**

- The **matplotlib** and **seaborn** libraries can be used to generate interesting visualizations in Python.
- The matplotlib library has excellent documentation on how to make different graphs <https://matplotlib.org/>.

Iris Data Set Example

- The Iris dataset is a classic dataset used for classification tasks in machine learning. It consists of 150 samples from three species of Iris flowers (Iris setosa, Iris versicolor, and Iris virginica), with four features measured for each sample: sepal length, sepal width, petal length, and petal width.

```
import matplotlib.pyplot as plt
import seaborn as sns
# Load the Iris dataset from seaborn
iris = sns.load_dataset('iris')
iris
```

This will load the data set as a **pandas** data frame, which is a powerful data structure for data manipulation and analysis.

	sepal_length	sepal_width	petal_length	petal_width	species
0	5.1	3.5	1.4	0.2	setosa
1	4.9	3.0	1.4	0.2	setosa
2	4.7	3.2	1.3	0.2	setosa
3	4.6	3.1	1.5	0.2	setosa
4	5.0	3.6	1.4	0.2	setosa
...	...	...	...	...	...
145	6.7	3.0	5.2	2.3	virginica
146	6.3	2.5	5.0	1.9	virginica
147	6.5	3.0	5.2	2.0	virginica
148	6.2	3.4	5.4	2.3	virginica
149	5.9	3.0	5.1	1.8	virginica

150 rows × 5 columns

You can use **pandas** to display basic statistics about the dataset:

```
# Display basic statistics about the dataset  
iris.describe()
```

	sepal_length	sepal_width	petal_length	petal_width
count	150.000000	150.000000	150.000000	150.000000
mean	5.843333	3.057333	3.758000	1.199333
std	0.828066	0.435866	1.765298	0.762238
min	4.300000	2.000000	1.000000	0.100000
25%	5.100000	2.800000	1.600000	0.300000
50%	5.800000	3.000000	4.350000	1.300000
75%	6.400000	3.300000	5.100000	1.800000
max	7.900000	4.400000	6.900000	2.500000

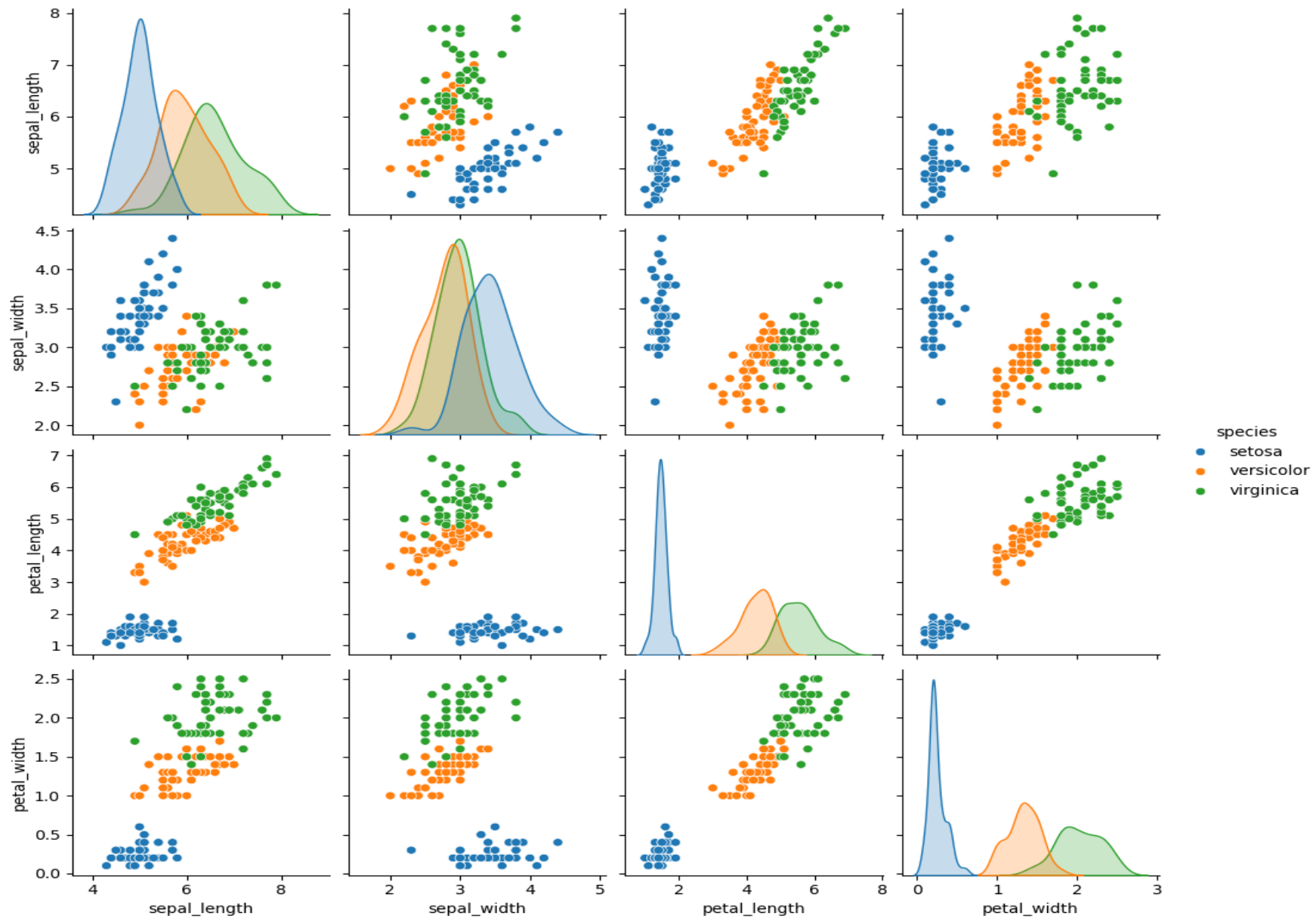
Display a pair plot of the features

A pairplot is a grid of scatter plots that allows us to visualize the relationships between different features in the dataset. The hue parameter is used to color the points based on the species of the Iris flower.

```
# Pairplot to visualize relationships between features
sns.pairplot(iris, hue='species')
plt.show()
```

This plot is useful to identify patterns and relationships between the features and the species. Notice that the diagonal plots are histograms or KDE plots of each feature for each species.

The KDE is a Kernel Density Estimate, which is a smooth, continuous representation of the distribution of a dataset.

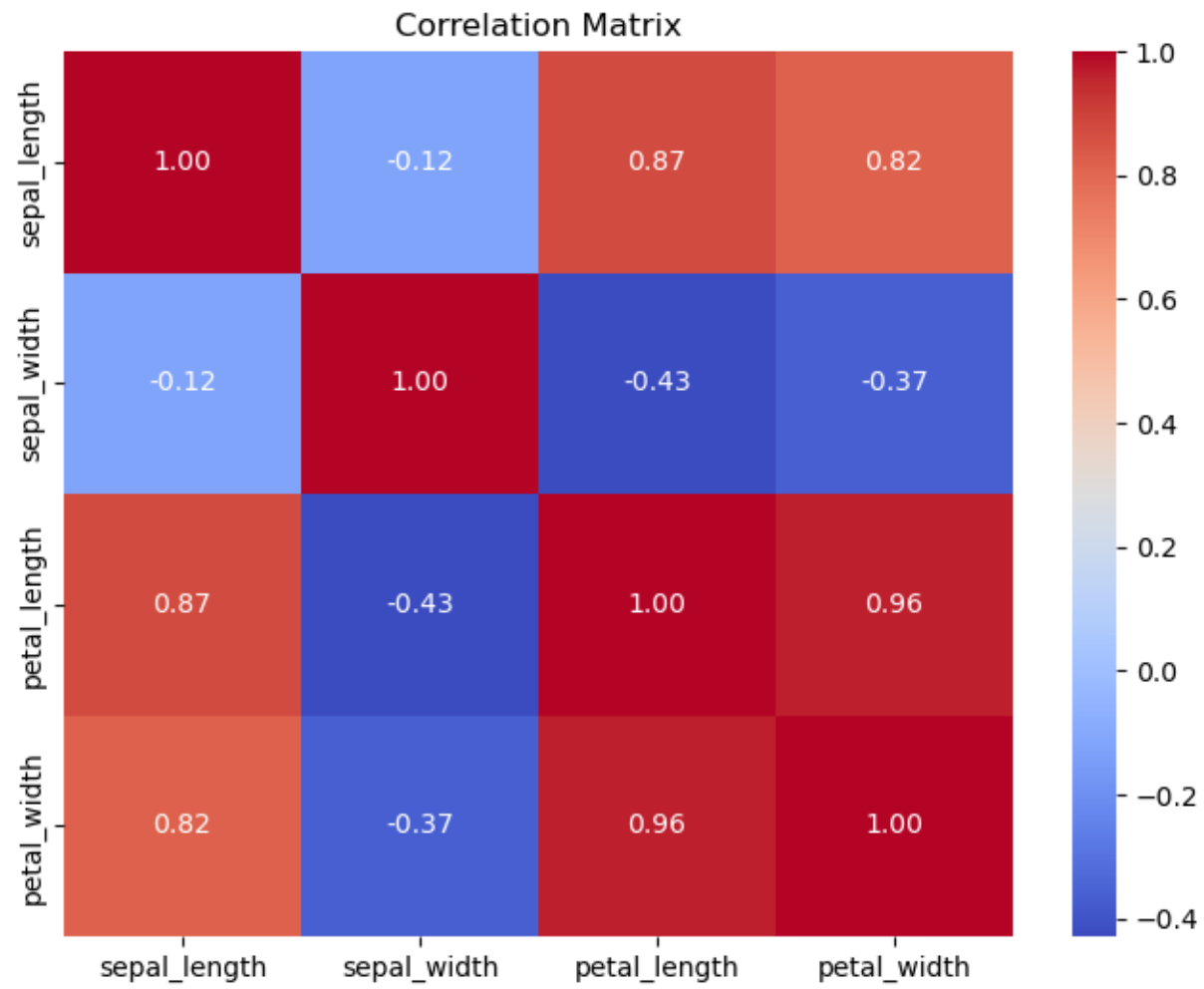


Heatmap of the correlation matrix

- The heatmap of the correlation matrix helps us understand the relationships between the features.
- The closer the value to 1, the stronger the positive correlation, and the closer to -1, the stronger the negative correlation. Values around 0 indicate no correlation.
- Notice that the diagonal elements are always 1 because the correlation of a variable with itself is always 1.

```
# Heatmap of the correlation matrix
# The iris DataFrame contains a non-numeric column (species)
# which cannot be included in a correlation matrix.
# The corr() function in pandas computes pairwise correlation of columns,
# excluding NA/null values,
# and it only works with numeric data.

plt.figure(figsize=(8, 6))
sns.heatmap(iris.drop('species', axis=1).corr(), annot=True, cmap='coolwarm', fmt=".2f")
plt.title('Correlation Matrix')
plt.show()
```



Interactive scatter plot

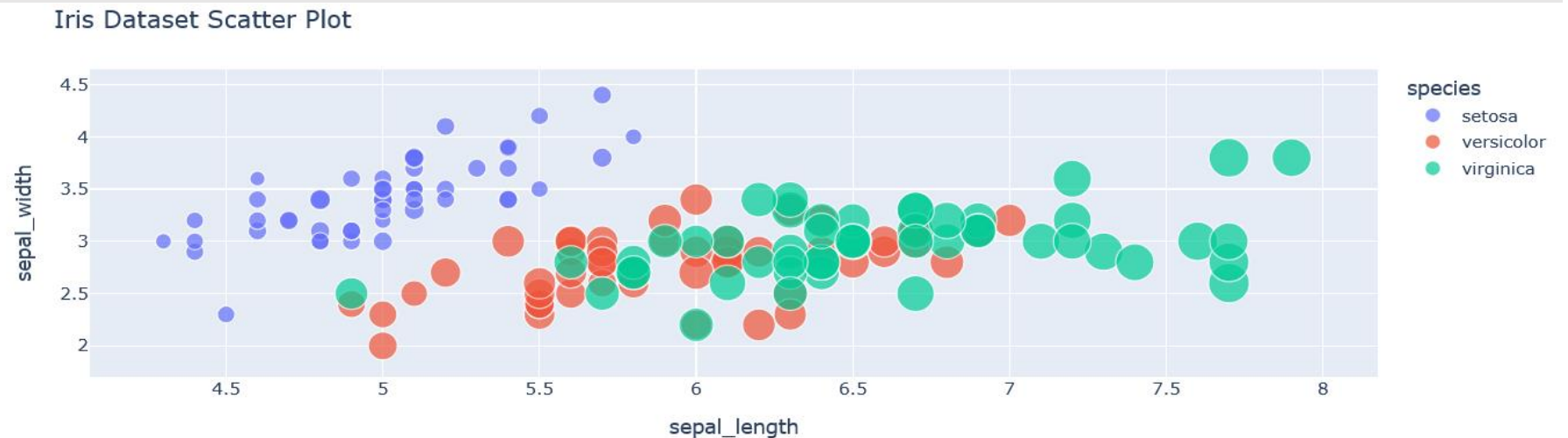
- Interactive scatter plot using **Plotly**

```
# Interactive scatter plot using Plotly
```

```
import plotly.express as px
```

```
fig = px.scatter(iris, x='sepal_length', y='sepal_width', color='species',  
size='petal_length', hover_data=['petal_width'])fig.update_layout(title='Iris Dataset  
Scatter Plot')
```

```
fig.show()
```



Review - Correlation

Correlation describes the linear relationship between two variables. It can be described as either strong or weak, and as either positive or negative.

$$\rho(X, Y) = \frac{\text{cov}(X, Y)}{\sqrt{\text{var}(X) \text{var}(Y)}} = \frac{\sigma_{xy}}{\sigma_x \sigma_y}$$

$$-1 \leq \rho(X, Y) \leq 1$$

